

Optimal Execution

Foundational Framework

As the foundational framework for optimal execution, we present the model of Almgren and Chriss [1]. An investor holding X shares must liquidate the full position by a fixed horizon T . Liquidating quickly concentrates trading into a short window and pushes execution prices away from the pre-trade price: **market impact**. Liquidating slowly reduces that cost but leaves the position exposed to the market's random drift for longer: **timing risk**. Almgren and Chriss cast this trade-off as a mean-variance optimization over the liquidation schedule.

Discretize the horizon into N steps of length $\tau = T/N$. Let x_k denote the shares still held after step k , with $x_0 = X$ and $x_N = 0$, and let $n_k = x_{k-1} - x_k$ be the shares traded in step k . Three parameters drive the cost: permanent impact γ (the lasting price shift per share traded, which the market never unwinds), temporary impact η (the transient price concession incurred for trading at a given rate, which decays once trading stops), and volatility σ (the size, in dollars per share, of the asset's random price moves per unit time).

Permanent impact contributes $\frac{1}{2}\gamma X^2$ to expected cost: each share traded shifts the price by a further increment of γ , so the average share pays half the full depression γX , giving a total of $X \cdot \frac{1}{2}\gamma X = \frac{1}{2}\gamma X^2$. This is fixed by the total size X and unaffected by the schedule, since every admissible schedule liquidates the same X . Temporary impact contributes $(\eta/\tau) \sum_{k=1}^N n_k^2$: because the cost of a step is quadratic in its trading rate, spreading the same shares over more time reduces it. The variance of total cost, driven by how many shares remain exposed and for how long, is $\sigma^2 \sum_{k=1}^N \tau x_k^2$. Almgren and Chriss minimize

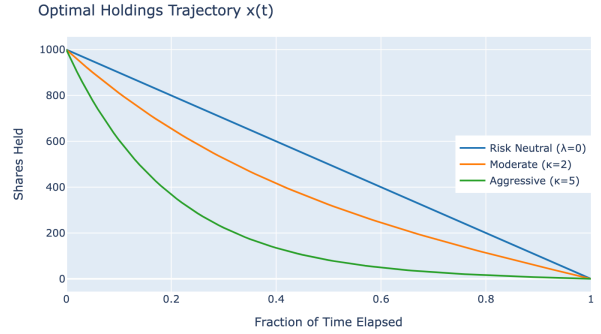
$$\mathbb{E}(\text{cost}) + \lambda \mathbb{V}(\text{cost})$$

over the schedule $x_1, \dots, x_{N-1}$, where $\lambda \geq 0$ is the trader's risk aversion. The full derivation, taken to its continuous-time limit, is given in the Appendix.

The optimal schedule is

$$x(t) = X \frac{\sinh(\kappa(T-t))}{\sinh(\kappa T)}, \quad \kappa = \sqrt{\lambda \sigma^2 / \eta}.$$

The single parameter κ governs the shape of the liquidation curve. As $\lambda \rightarrow 0$ (risk-neutral), $\kappa \rightarrow 0$ and the schedule becomes linear: a constant liquidation rate, i.e. minimum-impact, TWAP-like execution. As λ grows, κ grows and the curve front-loads: more of the position is sold early, trading higher impact cost for lower exposure to timing risk. The quantity $1/\kappa$ sets the characteristic time scale (the "half-life") over which the position is unwound.



Accumulating a position instead of liquidating one leaves this analysis unchanged: the objective is direction-blind, since cost enters only through n_k^2 , so only the boundary conditions flip, from $x_0 = X, x_N = 0$ to $x_0 = 0, x_N = X$. The optimal buy schedule is the exact time-reversal of the sell schedule, $x_{\text{buy}(t)} = X \sinh(\kappa t) / \sinh(\kappa T)$, obtained from the sell formula by swapping $t \leftrightarrow T - t$.

Temporary price impact is taken to be linear in the trading rate: executing at rate $\dot{x}(t)$ displaces the execution price by an amount proportional to $\dot{x}(t)$. Because the cost accrued over an instant is the product of this displacement and the amount traded, it is quadratic in the rate, consistent with the $\eta n_k^2 / \tau$ term above.

References

- [1] R. Almgren and N. Chriss, “Optimal execution of portfolio transactions,” *Journal of Risk*, vol. 3, no. 2, pp. 5–39, 2000.

Appendix

Notation

symbol	meaning	units
X	total shares to be liquidated (or accumulated)	shares
T	trading horizon	day
$\tau = T/N$	length of one discretization step, with N the number of steps	day
$x_k, x(t)$	shares still held after step k , or at time t	shares
$n_k = x_{k-1} - x_k$	shares traded in step k	shares
n_k/τ	trading rate during step k : shares traded per unit time, not to be confused with n_k itself	shares/day
γ	permanent-impact coefficient: the persistent price shift per share traded	\$/share ²
η	temporary-impact coefficient: the price concession per unit trading rate	\$/share ²
σ	volatility of the traded asset	\$/share/ $\sqrt{\text{day}}$
λ	risk aversion: the trader's chosen weight on cost variance	1/\$
$\kappa = \sqrt{\lambda\sigma^2/\eta}$	decay parameter setting the schedule's unwind timescale	1/day

γ and η each carry an extra factor of 1/ share beyond what “a price shift per share traded” suggests, because **price** is itself quoted in \$/share: a shift in that per-share price, per share traded, is (\$/share)/share = \$/share².

Almgren-Chriss derivation

Following [1], the discrete-time objective introduced in the Foundational Framework section is

$$\min_{x_1, \dots, x_{N-1}} \underbrace{\frac{1}{2}\gamma X^2 + \frac{\eta}{\tau} \sum_{k=1}^N n_k^2}_{\mathbb{E}[\text{cost}]} + \underbrace{\lambda\sigma^2 \sum_{k=1}^N \tau x_k^2}_{\mathbb{V}[\text{cost}]}.$$

The permanent-impact term $\frac{1}{2}\gamma X^2$ does not depend on the schedule $x_1, \dots, x_{N-1}$ and drops out of the minimization. Naming the remaining coefficients $a = \eta/\tau$ and $b = \lambda\sigma^2\tau$ turns what is left into

$$F = a \sum_{k=1}^N (x_{k-1} - x_k)^2 + b \sum_{k=1}^{N-1} x_k^2,$$

with the endpoints $x_0 = X$ and $x_N = 0$ held fixed and only the interior holdings $x_1, \dots, x_{N-1}$ free.

First-order condition. F is a sum of squares, a convex bowl with a single minimum where every partial derivative vanishes. For an interior x_j , the only terms containing it are $(x_{j-1} - x_j)^2$, $(x_j - x_{j+1})^2$, and $b x_j^2$, so

$$\frac{\partial F}{\partial x_j} = a[-2(x_{j-1} - x_j) + 2(x_j - x_{j+1})] + 2b x_j = 0.$$

Dividing by 2 and rearranging gives a recurrence linking each holding to its two neighbors:

$$x_{j+1} - 2x_j + x_{j-1} = \frac{b}{a} x_j = \frac{\lambda\sigma^2\tau^2}{\eta} x_j.$$

The recurrence is a discrete second derivative. The left-hand side is exactly the centered second-difference stencil, $x_{j+1} - 2x_j + x_{j-1} \approx \tau^2 \ddot{x}(t_j)$. Dividing the recurrence by τ^2 and taking $N \rightarrow \infty$ ($\tau \rightarrow 0$) turns it into the ODE

$$\ddot{x}(t) = \kappa^2 x(t), \quad \kappa^2 = \frac{\lambda \sigma^2}{\eta}.$$

The discrete first-order condition and the continuous Euler-Lagrange equation for this problem coincide; the recurrence above is simply that equation sampled on a grid.

Solving the ODE. The general solution of $\ddot{x} = \kappa^2 x$ is $x(t) = Ae^{\kappa t} + Be^{-\kappa t}$. For a **sell** schedule, impose $x(0) = X$ and $x(T) = 0$:

$$A + B = X, \quad Ae^{\kappa T} + Be^{-\kappa T} = 0.$$

Solving for A, B and folding the exponentials into hyperbolic sines ($\sinh z = \frac{1}{2}(e^z - e^{-z})$) collapses this to the single closed form

$$x(t) = X \frac{\sinh(\kappa(T-t))}{\sinh(\kappa T)}.$$

Risk-neutral limit. As $\kappa \rightarrow 0$ the expression is a 0/0 indeterminate; using $\sinh(z) \approx z$ for small z ,

$$\frac{\sinh(\kappa(T-t))}{\sinh(\kappa T)} \xrightarrow{\kappa \rightarrow 0} \frac{\kappa(T-t)}{\kappa T} = \frac{T-t}{T},$$

so $x(t) = X(1 - t/T)$: constant-rate liquidation, recovering the $\lambda = 0$ (pure minimum-impact) case.

Buying as the mirror image. Accumulating a position instead of liquidating one leaves the objective, and therefore the ODE, unchanged: n_k (and $\dot{x}$) enter only squared, so cost is direction-blind. Only the boundary conditions flip, to $x(0) = 0$ and $x(T) = X$, which gives $A = -B$ and

$$x(t) = X \frac{\sinh(\kappa t)}{\sinh(\kappa T)}.$$

The sell and buy schedules are exact time-reversals of one another, related by swapping $t \leftrightarrow T - t$:

$$x_{\text{sell}(t)} = X \frac{\sinh(\kappa(T-t))}{\sinh(\kappa T)}, \quad x_{\text{buy}(t)} = X \frac{\sinh(\kappa t)}{\sinh(\kappa T)}.$$